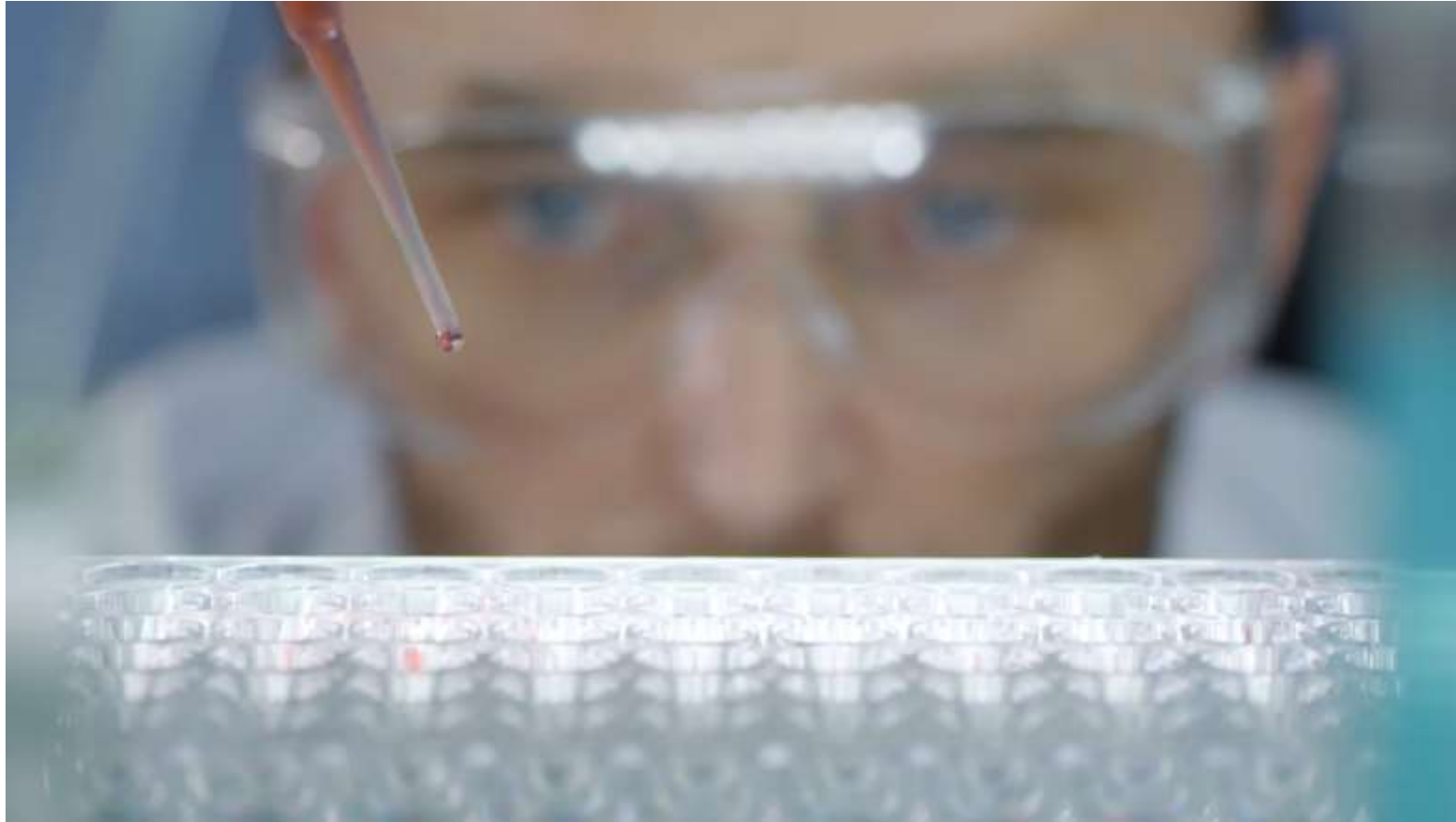




Digital Forensics Project

By: Oluwatoni Oderinlo





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- Executive Summary
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- Conclusion



Executive Summary

Security Breach: Unauthorized access and data exfiltration

Date of Incident: September 18, 2020.

Entry Vector: Remote Desktop Protocol (RDP)

Malicious Activity: Deployment of coreupdater.exe

Stolen Files: CA root certificate, Szechuan sauce recipe

Attacker's IP: 194.61.24.102



Technical Findings



When was the system
compromised?



Date: September 18, 2020, at
10:24 PM



Initial Entry Vector:
Unauthorized RDP
connection



84361	15679.078822	194.61.24.102	10.42.85.10	RDP	108 Cookie: mstshash=nmap, Negotiate Request
84369	15679.322087	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2304...	15798.642147	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2304...	15798.644463	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2304...	15798.872140	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2304...	15798.873905	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2304...	15799.094343	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2304...	15799.096253	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2304...	15799.316468	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2304...	15799.318221	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2304...	15799.531085	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2305...	15799.532750	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2305...	15799.754182	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2305...	15799.755939	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2305...	15799.967178	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2305...	15799.968991	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2305...	15800.191349	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2305...	15800.193286	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2305...	15800.417256	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2305...	15800.418976	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2306...	15800.628706	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2306...	15800.630257	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2306...	15800.851693	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2306...	15800.853473	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2306...	15801.076409	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2306...	15801.078346	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2306...	15801.291203	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2306...	15801.292961	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2306...	15801.513587	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2306...	15801.515257	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2307...	15801.727310	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request
2307...	15801.729046	10.42.85.10	194.61.24.102	RDP	85 Negotiate Response
2307...	15801.949357	194.61.24.102	10.42.85.10	RDP	117 Cookie: mstshash=Administrator, Negotiate Request

```

Ethernet II, Src: VMware_95:cd:21 (00:0c:29:95:cd:21), Dst: VMware_e1:84:e6 (00:0c:29:e1:84:e6)
Internet Protocol Version 4, Src: 194.61.24.102, Dst: 10.42.85.10
Transmission Control Protocol, Src Port: 38100, Dst Port: 3389, Seq: 1, Ack: 1, Len: 42
TPKT, Version: 3, Length: 42
ISO 8073/X.224 COTP Connection-Oriented Transport Protocol
Remote Desktop Protocol
  Routing Token/Cookie: Cookie: msthash=nmap
  Type: RDP Negotiation Request (0x01)
  Flags: 0x00
  Length: 8
  requestedProtocols: 0x0000000b, TLS security supported, CredSSP supported, CredSSP with Early User Authorization Res...

```

```

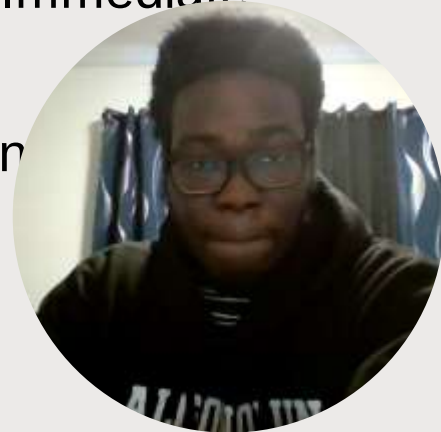
0000 00 0c 29 e1 84 e6 00 0c 29 95 cd 21 08 00 45
0010 00 5e e6 8b 40 00 3f 06 1b 37 c2 3d 18 66 0a
0020 55 0a 94 d4 0d 3d ce 32 83 6b 0f 25 3d 7a 8
0030 01 f6 6e 9c 00 00 01 01 08 0a b1 65 fc 17 0
0040 b3 07 03 00 00 2a 25 e0 00 00 00 00 43 6
0050 6b 69 65 3a 20 6d 73 74 73 68 61 73 68 3d 6
0060 61 70 0d 0a 01 00 08 00 0b 00 00 00

```



How was it compromised?

1. Initial entry vector: RDP exploitation
2. Utilization of Malicious Software: coreupdater.exe
3. Persistence and Network Exploration: Lateral movement within the network
4. Data Compromise: Theft of sensitive files
5. Network Setup and Security Enhancements: Immediate actions for network security
6. Final Communication with the Adversary: Transfer of encrypted application data





Who compromised the system?

- Attacking IP: 194.61.24.102
- Malicious Actions:
 - Unauthorized access to sensitive files
 - Introduction of coreupdater.exe malware
 - Persistence on compromised systems
 - Lateral movement within the network
 - Data theft from both server and desktop



case001.pcap

FileEditViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

http

No.	Time	Source	Destination	Protocol	Length	Info
84086	15512.939062	10.42.85.115	104.126.211.229	HTTP/X...	262	POST /fwlink/?LinkID=252669&clid=0x409 HTTP/1.1
84089	15512.954074	104.126.211.229	10.42.85.115	HTTP	350	HTTP/1.1 302 Moved Temporarily
84103	15513.243952	10.42.85.115	20.54.64.202	HTTP/X...	262	POST /metadata.svc HTTP/1.1
84108	15513.503879	20.54.64.202	10.42.85.115	HTTP/X...	692	HTTP/1.1 200 OK
84142	15526.944949	10.42.85.115	72.21.91.29	HTTP	296	GET /MFEwTzBNMEswSTAJBgUrDgMCGGUABBBQ50otx%2Fh0zt1%2Bz8SiPI7wEWVxDlQQUTiJUIBiV5uNu5g%2F6%2Brks7QYXjzkCEALnkXH7gChpP%2BLZg4NMUMA%...
84144	15526.964237	72.21.91.29	10.42.85.115	OCSP	853	Response
2360...	15902.183653	10.42.85.10	204.79.197.200	HTTP	274	GET /favicon.ico HTTP/1.1
2360...	15902.221545	204.79.197.200	10.42.85.10	HTTP	620	HTTP/1.1 200 OK (PNG)
2361...	15902.619926	10.42.85.10	72.21.91.29	HTTP	289	GET /MFEwTzBNMEswSTAJBgUrDgMCGGUABBBTL0V27RVZ7LBduom%2FnYB45SPUEwQU5Z1ZMIJHWMys%2BghUNoZ70rUETfACEAiIzVJfGSRETRSlgpHeuVI%3D HTTP/1.1
2361...	15902.641772	72.21.91.29	10.42.85.10	OCSP	430	Response
2361...	15902.879869	10.42.85.10	72.21.91.29	HTTP	291	GET /MFEwTzBNMEswSTAJBgUrDgMCGGUABBBTL0V27RVZ7LBduom%2FnYB45SPUEwQU5Z1ZMIJHWMys%2BghUNoZ70rUETfACEA8sEM1bBsCTf7jUSfg%2Bhwk%3D HTTP/1.1
2361...	15902.894998	72.21.91.29	10.42.85.10	OCSP	429	Response
2366...	15927.110514	10.42.85.10	23.57.58.91	HTTP	284	GET /browserconfig.xml HTTP/1.1
2366...	15927.162685	23.57.58.91	10.42.85.10	HTTP	1504	HTTP/1.1 404 Not Found (text/html)
2367...	15934.261595	10.42.85.10	194.61.24.102	HTTP	302	GET / HTTP/1.1
2367...	15934.264353	194.61.24.102	10.42.85.10	HTTP	420	HTTP/1.0 200 OK (text/html)
2368...	15934.326800	10.42.85.10	194.61.24.102	HTTP	255	GET /favicon.ico HTTP/1.1
2368...	15934.327590	194.61.24.102	10.42.85.10	HTTP	370	HTTP/1.0 404 File not found (text/html)
2385...	15959.468916	10.42.85.10	194.61.24.102	HTTP	291	GET /coreupdater.exe HTTP/1.1
2385...	15959.469636	194.61.24.102	10.42.85.10	HTTP	110	HTTP/1.0 200 OK (application/x-msdos-program)
2658...	16696.836744	10.42.85.115	23.57.58.91	HTTP/X...	1472	POST /fwlink/?LinkID=252669&clid=0x409 HTTP/1.1
2658...	16696.885023	23.57.58.91	10.42.85.115	HTTP	350	HTTP/1.1 302 Moved Temporarily
2659...	16697.185625	10.42.85.115	20.189.118.208	HTTP/X...	1472	POST /metadata.svc HTTP/1.1
2663...	16697.558423	20.189.118.208	10.42.85.115	HTTP/X...	692	HTTP/1.1 200 OK
2664...	16697.625607	10.42.85.115	23.57.58.91	HTTP/X...	652	POST /fwlink/?LinkID=252669&clid=0x409 HTTP/1.1
2664...	16697.676328	23.57.58.91	10.42.85.115	HTTP	350	HTTP/1.1 302 Moved Temporarily
2665...	16697.684072	10.42.85.115	20.189.118.208	HTTP/X...	652	POST /metadata.svc HTTP/1.1
2667...	16698.048122	20.189.118.208	10.42.85.115	HTTP/X...	692	HTTP/1.1 200 OK
2667...	16698.101737	10.42.85.115	23.57.58.91	HTTP/X...	1474	POST /fwlink/?LinkID=252669&clid=0x409 HTTP/1.1

Frame 238574: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface unknown, id 0

Ethernet II, Src: VMware_95:cd:21 (00:0c:29:95:cd:21), Dst: VMware_e1:84:e6 (00:0c:29:e1:84:e6)

Internet Protocol Version 4, Src: 194.61.24.102, Dst: 10.42.85.10

Transmission Control Protocol, Src Port: 80, Dst Port: 62410, Seq: 7318, Ack: 238, Len: 56

[7 Reassembled TCP Segments (7373 bytes): #238567(17), #238569(1460), #238570(1460), #238571(1460), #238572(1460), #238573(1460), #238574(110)]

Hypertext Transfer Protocol

HTTP/1.0 200 OK\r\n

Server: SimpleHTTP/0.6 Python/2.7.18\r\n

Date: Sat, 19 Sep 2020 02:24:06 GMT\r\n

Content-type: application/x-msdos-program\r\n

Content-Length: 7168\r\n

Last-Modified: Fri, 18 Sep 2020 23:29:15 GMT\r\n

\r\n

[HTTP response 1/1]

0000 00 0c 29 e1 84 e6 00 0c 29 95 cd 21 08 00 45 00

0010 00 60 5e 89 40 00 3f 06 a3 37 c2 3d 18 66 0a 2a

0020 55 0a 00 50 f3 ca 5c e8 c7 30 67 70 66 84 50 18

0030 01 f5 8d 9f 00 00 00 00 00 00 00 00 00 00 00

0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Frame (110 bytes)

Reassembled TCP (7373 bytes)

Packets: 411797 · Displayed: 456 (0.1%)

Profile: Default

Filesystem tree view showing folders like Secret (7), PerfLogs (2), Program Files (11), Program Files (x86) (9), ProgramData (12), System Volume Information (5), Users (8), Windows (86), ADFS (39), ADWS (8), AppCompat (3), apppatch (15), AppReadiness (2), assembly (8), Boot (6), Branding (4), CbsTemp (2), Cursors (118), debug (21), DesktopTileResources (5), diagnostics (4), DigitalLocker (3), Downloaded Program Files (3), drivers (2), ELAMBKUP (2), en-US (9), Fonts (475), Globalization (5), Help (8), IME (7), ImmersiveControlPanel (3), Inf (694), InputMethod (7), Installer (8), L2Schemas (10), LiveKernelReports (2), Logs (7), media (86), Microsoft.NET (6), ModemLogs (2), NTDS (10), Offline Web Pages (3), Panther (16), PLA (6), PolicyDefinitions (185), Provisioning (3), Registration (6), RemotePackages (4), rescache (3), Resources (4), SchCache (2).

Keyword search

Table Thumbnail Summary

Save Table as CSV

Name	Keyword Preview	Location	Modified Time	Change Time	Access Time	Created Time	Size	Flag
shell32.dll.mui	g;updater;up-dater;«updater.exe»;updates;up-dates;up	/img_20200918_0347_CDrive.E01/vol_vol3/Windows/Syst...	2014-03-21 14:49:16 EDT	2020-09-17 12:48:51 EDT	2014-03-21 14:49:16 EDT	2014-03-21 14:49:16 EDT	417280	Alloc
sysmain.sdb	API library module«updater.exe»UpdateWizard.exeUS	/img_20200918_0347_CDrive.E01/vol_vol3/Windows/app...	2014-03-21 14:49:25 EDT	2020-09-17 12:48:00 EDT	2014-03-21 14:49:25 EDT	2014-03-21 14:49:25 EDT	431006	Alloc
sysmain.sdb	DBE.exeP2P Barrier«Updater.exe»DbgCLR.exeVisual Studio	/img_20200918_0347_CDrive.E01/vol_vol3/Windows/app...	2014-03-21 14:48:57 EDT	2020-09-17 12:48:00 EDT	2014-03-21 14:48:56 EDT	2014-03-21 14:48:56 EDT	3828554	Alloc
sysmain.sdb	API library module«updater.exe»UpdateWizard.exeUS	/img_20200918_0347_CDrive.E01/vol_vol3/Windows/WinS...	2014-03-21 14:49:25 EDT	2020-09-17 12:48:00 EDT	2014-03-21 14:49:25 EDT	2014-03-21 14:49:25 EDT	431006	Alloc
shell32.dll.mui	g;updater;up-dater;«updater.exe»;updates;up-dates;up	/img_20200918_0347_CDrive.E01/vol_vol3/Windows/Sys...	2014-03-21 14:49:16 EDT	2020-09-17 12:48:51 EDT	2014-03-21 14:49:16 EDT	2014-03-21 14:49:16 EDT	417280	Alloc

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Indexed Text Translation

Page: 11 of 27 Page Matches on page: 1 of 1 Match 100% Reset

Updater.exe
DbgCLR.exe
Visual Studio .NET
Microsoft
Visual Studio*
DISABLEWIN9XMODE;VIRTUALIZEHKCR
SamplingControl.dll
SHLWAPI.dll
Microsoft
Visual Studio .NET
Microsoft CLR Debugger
dbgout.exe



What did they do?

- Unauthorized Access: Accessed sensitive files, including CA root certificate and Szechuan sauce recipe
- Introduction of Malicious Software: Deployed coreupdater.exe malware
- Access to Folders/Files: Yes, accessed sensitive files
- Use of Specific Tools: Yes, utilized coreupdater.exe
- Execution of Programs/Scripts: Yes, executed malicious programs



cap

ViewGoCaptureAnalyzeStatisticsTelephonyWirelessToolsHelp

=194.61.24.102

Time	Source	Destination	Protocol	Length	Info
5798.860479	10.42.85.10	194.61.24.102	TCP	60	3389 → 40044 [RST, ACK] Seq=1728 Ack=1683 Win=0 Len=0
5798.871395	194.61.24.102	10.42.85.10	TCP	74	[TCP Port numbers reused] 40046 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2976370696 TSecr=0 WS=128
5798.871857	10.42.85.10	194.61.24.102	TCP	74	3389 → 40046 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM TSval=713175 TSecr=2976370696
5798.872112	194.61.24.102	10.42.85.10	TCP	66	40046 → 3389 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2976370697 TSecr=713175
5798.872140	194.61.24.102	10.42.85.10	RDP	117	Cookie: msthash=Administrator, Negotiate Request
5798.873905	10.42.85.10	194.61.24.102	RDP	85	Negotiate Response
5798.874099	194.61.24.102	10.42.85.10	TCP	66	40046 → 3389 [ACK] Seq=52 Ack=20 Win=64256 Len=0 TSval=2976370699 TSecr=713175
5798.874346	194.61.24.102	10.42.85.10	TLSv1.2	379	Client Hello (SNI=10.42.85.10)
5798.875555	10.42.85.10	194.61.24.102	TLSv1.2	1257	Server Hello, Certificate, Server Key Exchange, Server Hello Done
5798.875733	194.61.24.102	10.42.85.10	TCP	66	40046 → 3389 [ACK] Seq=365 Ack=1211 Win=64128 Len=0 TSval=2976370700 TSecr=713175
5798.876010	194.61.24.102	10.42.85.10	TLSv1.2	248	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
5798.876991	10.42.85.10	194.61.24.102	TLSv1.2	173	Change Cipher Spec, Encrypted Handshake Message
5798.877183	194.61.24.102	10.42.85.10	TCP	66	40046 → 3389 [ACK] Seq=547 Ack=1318 Win=64128 Len=0 TSval=2976370702 TSecr=713175
5798.877453	194.61.24.102	10.42.85.10	TLSv1.2	231	Application Data
5798.877838	10.42.85.10	194.61.24.102	TLSv1.2	391	Application Data
5798.878063	194.61.24.102	10.42.85.10	TCP	66	40046 → 3389 [ACK] Seq=712 Ack=1643 Win=64128 Len=0 TSval=2976370703 TSecr=713175
5798.981684	194.61.24.102	10.42.85.10	TLSv1.2	951	Application Data
5798.983009	10.42.85.10	194.61.24.102	TLSv1.2	151	Application Data
5798.983230	194.61.24.102	10.42.85.10	TCP	66	40046 → 3389 [ACK] Seq=1597 Ack=1728 Win=64128 Len=0 TSval=2976370808 TSecr=713186
5799.085052	194.61.24.102	10.42.85.10	TLSv1.2	151	Encrypted Alert
5799.085089	194.61.24.102	10.42.85.10	TCP	66	40046 → 3389 [FIN, ACK] Seq=1682 Ack=1728 Win=64128 Len=0 TSval=2976370910 TSecr=713186
5799.085297	10.42.85.10	194.61.24.102	TCP	66	3389 → 40046 [ACK] Seq=1728 Ack=1683 Win=63915 Len=0 TSval=713197 TSecr=2976370909
5799.085347	10.42.85.10	194.61.24.102	TCP	60	3389 → 40046 [RST, ACK] Seq=1728 Ack=1683 Win=0 Len=0
5799.093790	194.61.24.102	10.42.85.10	TCP	74	[TCP Port numbers reused] 40048 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2976370918 TSecr=0 WS=128
5799.094061	10.42.85.10	194.61.24.102	TCP	74	3389 → 40048 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 WS=1 SACK_PERM TSval=713197 TSecr=2976370918
5799.094313	194.61.24.102	10.42.85.10	TCP	66	40048 → 3389 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2976370919 TSecr=713197
5799.094343	194.61.24.102	10.42.85.10	RDP	117	Cookie: msthash=Administrator, Negotiate Request
5799.096253	10.42.85.10	194.61.24.102	RDP	85	Negotiate Response
5799.096486	194.61.24.102	10.42.85.10	TCP	66	40048 → 3389 [ACK] Seq=52 Ack=20 Win=64256 Len=0 TSval=2976370921 TSecr=713198
5799.096699	194.61.24.102	10.42.85.10	TLSv1.2	379	Client Hello (SNI=10.42.85.10)
5799.096750	10.42.85.10	194.61.24.102	TLSv1.2	1257	Server Hello, Certificate, Server Key Exchange, Server Hello Done

30439: 248 bytes on wire (1984 bits), 248 bytes captured (1984 bits) on interface unknown, id 0

II, Src: VMware_95:cd:21 (00:0c:29:95:cd:21), Dst: VMware_e1:84:e6 (00:0c:29:e1:84:e6)

Protocol Version 4, Src: 194.61.24.102, Dst: 10.42.85.10

Session Control Protocol, Src Port: 40046, Dst Port: 3389, Seq: 365, Ack: 1211, Len: 182

Record Layer Security

1.2 Record Layer: Handshake Protocol: Client Key Exchange

Content Type: Handshake (22)

Version: TLS 1.2 (0x0303)

Length: 70

Handshake Protocol: Client Key Exchange

1.2 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec

1.2 Record Layer: Handshake Protocol: Encrypted Handshake Message

0000 00 0c 29 e1 84 e6 00 0c 29 95 cd 21 08 00 45 00

0010 00 ea dc 80 40 00 3f 06 24 b6 c2 3d 18 66 0a 2

0020 55 0a 9c 6e 0d 3d 79 11 7b 70 62 07 02 bb 80 7

0030 01 f5 e0 b0 00 00 01 01 08 0a b1 67 d0 0c 00

0040 e1 d7 16 03 03 00 46 10 00 00 42 41 04 2f eb

0050 8c 2d f3 f0 2f 27 49 de c2 1c af 45 aa 24 12

0060 3c 79 ce cb 2a e3 28 7b 14 9a dc 4d a3 44 35

0070 4b 4e f0 d4 56 6f 54 e2 5c e9 bc aa 38 96 37 8

0080 47 0f cf a7 92 73 b1 90 c1 cb cd 5b 7d 14 03 0

0090 00 01 01 16 03 03 00 60 5c d0 45 7b 30 bf 14 f1

00a0 57 9d 0d bd 3d 4a db 14 74 2a 21 9f 98 8d bf a0

00b0 69 ad 8e c1 e0 89 96 0d 21 5a 82 9a 8c b3 83 15

00c0 39 c9 fb 57 ed 79 70 b7 15 c6 0a ec 32 27 85 8a

00d0 ea dd fb 17 29 e3 47 96 d9 5d 01 1d 36 df fb b5






[illegible]

0050	0b	41	5b	20	39	93	ee	94	6b	08	a7	80	bc	f3	3
0060	c3	c7	3d	b4	40	69	88	09	4a	56	b7	57	72	42	a
0070	e7	47	cd	39	72	48	eb	8a	dd	0c	e9	bd	5b	91	5
0080	aa	75	21	3a	a1	4a	c4	3b	90	7f	f3	e4	31	ca	87
0090	eb	39	f5	12	28	65	52								



http

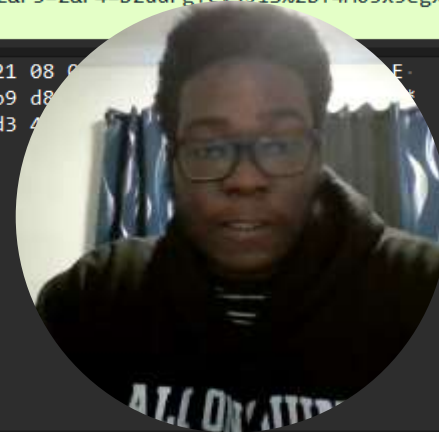
No.	Time	Source	Destination	Protocol	Length	Info
4097...	26742.333620	23.37.117.182	10.42.85.115	HTTP	333	HTTP/1.1 200 OK
4097...	26743.089728	10.42.85.115	205.185.216.10	HTTP	370	GET /filestreamingservice/files/3b79c97b-09ea-41fb-92f1-4589f4e2f3c3/pieceshash HTTP/1.1
4097...	26743.107285	205.185.216.10	10.42.85.115	HTTP	252	HTTP/1.1 200 OK
4098...	26743.142449	10.42.85.115	8.252.174.254	HTTP	475	GET /filestreamingservice/files/3b79c97b-09ea-41fb-92f1-4589f4e2f3c3?P1=1600493649&P2=402&P3=2&P4=dybMOALUBpuQjbzcgON2xBkv8Er4
4098...	26743.144242	10.42.85.115	8.253.200.120	HTTP	475	GET /filestreamingservice/files/3b79c97b-09ea-41fb-92f1-4589f4e2f3c3?P1=1600493649&P2=402&P3=2&P4=dybMOALUBpuQjbzcgON2xBkv8Er4
4098...	26743.162249	8.252.174.254	10.42.85.115	HTTP	1059	HTTP/1.1 206 Partial Content
4098...	26743.162960	8.253.200.120	10.42.85.115	HTTP	1059	HTTP/1.1 206 Partial Content
4099...	26743.748409	10.42.85.115	205.185.216.10	HTTP	370	GET /filestreamingservice/files/f7ded4c4-f468-4bdc-96f0-03fd60c5ae81/pieceshash HTTP/1.1
4099...	26743.767827	205.185.216.10	10.42.85.115	HTTP	252	HTTP/1.1 200 OK
4099...	26743.966115	10.42.85.115	104.124.63.145	HTTP	479	GET /filestreamingservice/files/f7ded4c4-f468-4bdc-96f0-03fd60c5ae81?P1=1600493649&P2=402&P3=2&P4=D2udPgTcv5JiS%2bY4M03x5egxGI
4099...	26743.966215	10.42.85.115	104.124.63.155	HTTP	479	GET /filestreamingservice/files/f7ded4c4-f468-4bdc-96f0-03fd60c5ae81?P1=1600493649&P2=402&P3=2&P4=D2udPgTcv5JiS%2bY4M03x5egxGI
4099...	26744.000829	104.124.63.155	10.42.85.115	HTTP	1006	HTTP/1.1 206 Partial Content
4099...	26744.004613	104.124.63.145	10.42.85.115	HTTP	1006	HTTP/1.1 206 Partial Content
4100...	26744.347273	10.42.85.115	8.252.174.254	HTTP	480	GET /filestreamingservice/files/3b79c97b-09ea-41fb-92f1-4589f4e2f3c3?P1=1600493649&P2=402&P3=2&P4=dybMOALUBpuQjbzcgON2xBkv8Er4
4101...	26744.528655	10.42.85.115	104.124.63.155	HTTP	484	GET /filestreamingservice/files/f7ded4c4-f468-4bdc-96f0-03fd60c5ae81?P1=1600493649&P2=402&P3=2&P4=D2udPgTcv5JiS%2bY4M03x5egxGI
4103...	26744.608529	8.252.174.254	10.42.85.115	HTTP	1204	HTTP/1.1 206 Partial Content
4104...	26744.697494	104.124.63.155	10.42.85.115	HTTP	963	HTTP/1.1 206 Partial Content
4105...	26746.044812	10.42.85.115	205.185.216.42	HTTP	477	GET /filestreamingservice/files/3b79c97b-09ea-41fb-92f1-4589f4e2f3c3?P1=1600493649&P2=402&P3=2&P4=dybMOALUBpuQjbzcgON2xBkv8Er4
4105...	26746.044839	10.42.85.115	205.185.216.10	HTTP	477	GET /filestreamingservice/files/3b79c97b-09ea-41fb-92f1-4589f4e2f3c3?P1=1600493649&P2=402&P3=2&P4=dybMOALUBpuQjbzcgON2xBkv8Er4
4105...	26746.061106	205.185.216.10	10.42.85.115	HTTP	426	HTTP/1.1 206 Partial Content
4105...	26746.061123	205.185.216.42	10.42.85.115	HTTP	56	HTTP/1.1 206 Partial Content
4105...	26746.737159	10.42.85.115	8.252.107.126	HTTP	475	GET /filestreamingservice/files/3b79c97b-09ea-41fb-92f1-4589f4e2f3c3?P1=1600493649&P2=402&P3=2&P4=dybMOALUBpuQjbzcgON2xBkv8Er4
4105...	26746.743082	10.42.85.115	8.253.203.126	HTTP	475	GET /filestreamingservice/files/3b79c97b-09ea-41fb-92f1-4589f4e2f3c3?P1=1600493649&P2=402&P3=2&P4=dybMOALUBpuQjbzcgON2xBkv8Er4
4105...	26746.759013	8.253.203.126	10.42.85.115	HTTP	1059	HTTP/1.1 206 Partial Content
4105...	26746.759060	8.252.107.126	10.42.85.115	HTTP	1059	HTTP/1.1 206 Partial Content
4105...	26746.891756	10.42.85.115	8.253.203.126	HTTP	480	GET /filestreamingservice/files/3b79c97b-09ea-41fb-92f1-4589f4e2f3c3?P1=1600493649&P2=402&P3=2&P4=dybMOALUBpuQjbzcgON2xBkv8Er4
4107...	26747.017267	8.253.203.126	10.42.85.115	HTTP	1204	HTTP/1.1 206 Partial Content
4107...	26749.599894	10.42.85.115	104.124.63.155	HTTP	484	GET /filestreamingservice/files/f7ded4c4-f468-4bdc-96f0-03fd60c5ae81?P1=1600493649&P2=402&P3=2&P4=D2udPgTcv5JiS%2bY4M03x5egxGI
4110...	26749.837999	104.124.63.155	10.42.85.115	HTTP	963	HTTP/1.1 206 Partial Content

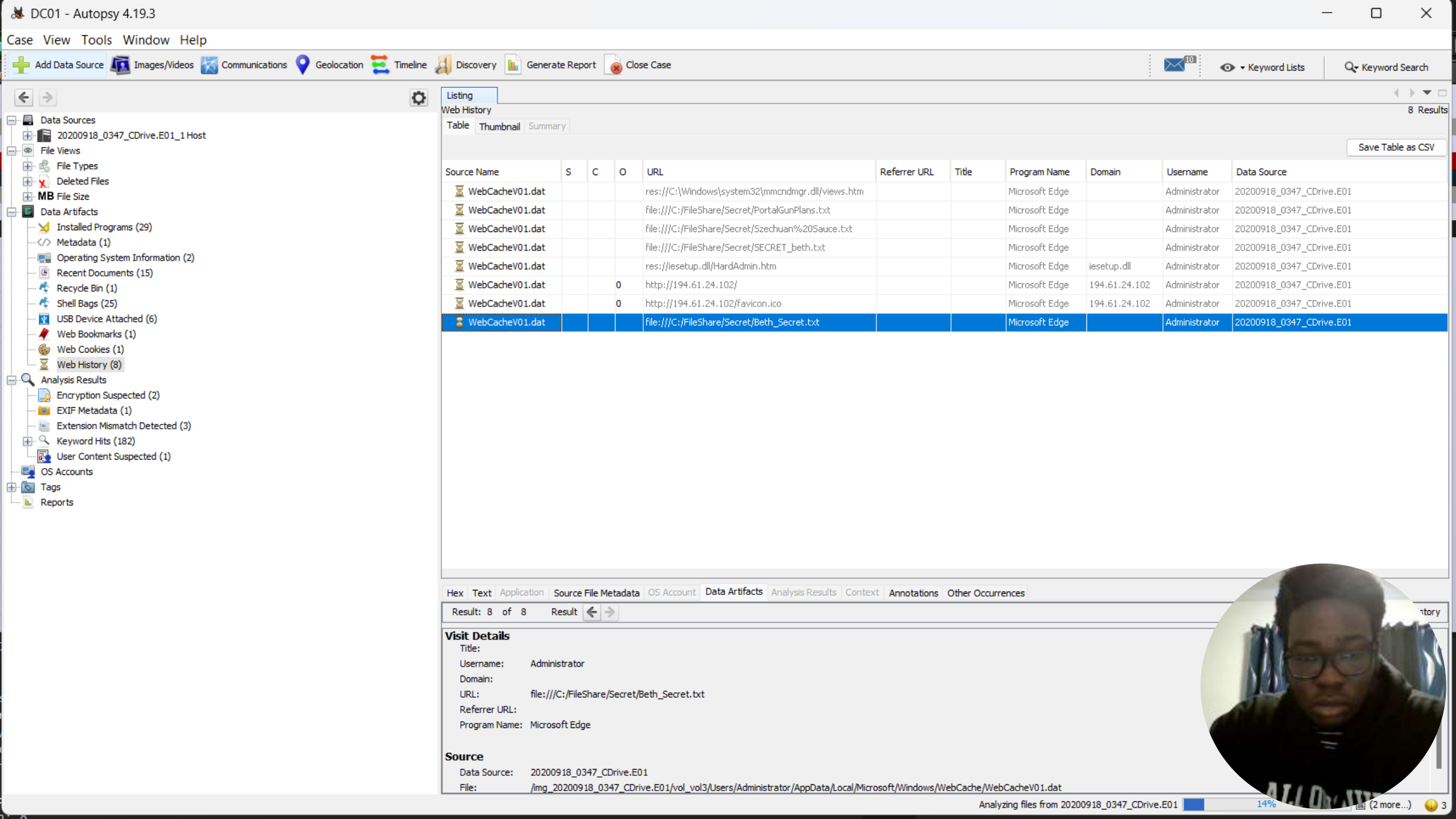
▶ Frame 410523: 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface unknown, id 0
▶ Ethernet II, Src: VMware_95:cd:21 (00:0c:29:95:cd:21), Dst: VMware_14:c2:95 (00:0c:29:14:c2:95)
▶ Internet Protocol Version 4, Src: 205.185.216.42, Dst: 10.42.85.115
▶ Transmission Control Protocol, Src Port: 80, Dst Port: 51173, Seq: 371, Ack: 424, Len: 2
▶ [2 Reassembled TCP Segments (372 bytes): #410521(370), #410523(2)]

Hypertext Transfer Protocol

▶ HTTP/1.1 206 Partial Content\r\n
Date: Sat, 19 Sep 2020 05:23:53 GMT\r\n
Connection: Keep-Alive\r\n
Cache-Control: public, max-age=1728000\r\n
Content-Length: 2\r\n
Content-Range: bytes 0-1/235196\r\n
Content-Type: application/octet-stream\r\n
Last-Modified: Wed, 30 Jan 2019 14:28:15 GMT\r\n

0000 00 0c 29 14 c2 95 00 0c 29 95 cd 21 08 00 00 00
0010 00 2a 94 7f 00 00 7f 06 a1 cd cd b9 d8 00 00 00
0020 55 73 00 50 c7 e5 5f 2d 6e 23 12 d3 40 00 00 00
0030 fa f0 67 ab 00 00 50 4b





Conclusion



Troubling security breach
resulting in unauthorized
access and data exfiltration



Initial entry via RDP
exploitation, followed by
deployment of
coreupdater.exe



Immediate actions required
to enhance network security
and revoke compromised CA
root certificate





Thank you!

